Folding scheme in SALSAA

0. Intuition

Lattice-based folding follows a similar methodology:

- start from linear relations
- norm check
- (linear combination) fold $\to$ decompose
  - or decompose $\to$ fold

1. Linear relation

1. **Row-tensor structure.**

- $\mathbf{F}$ can be written as $\mathbf{F} = \mathbf{F}_0 \bullet \ldots \bullet \mathbf{F}_{\mu-1}$, where $\mathbf{F}_i \in \mathcal{R}_q^{n\times d}$ for $i \in [\mu]$ and $\bullet$ denotes the row-wise tensor product, by the shorthand $\mathbf{F} \in \mathcal{R}_q^{n\times d^{\otimes\mu}}$.

Concretely, the row-wise tensor product means that for each row index $j \in [n]$, the j-th row of $\mathbf{F}$ is the tensor product of the j-th rows of $\mathbf{F}_0, \ldots, \mathbf{F}_{\mu-1}$.

- For example

$$\mathbf{F}_0 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \mathbf{F}_1 = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

  Then $\mathbf{F} = \mathbf{F}_0 \bullet \mathbf{F}_1 \in \mathcal{R}_q^{2\times 4}$ is obtained row by row:

$$\begin{aligned} \mathbf{f}_0 &= (1,2) \otimes (5,6) = (5,6,10,12), \\ \mathbf{f}_1 &= (3,4) \otimes (7,8) = (21,24,28,32). \end{aligned}$$

$$\mathbf{F} = \begin{bmatrix} \mathbf{f}_0 \\ \mathbf{f}_1 \end{bmatrix} = \begin{bmatrix} 5 & 6 & 10 & 12 \\ 21 & 24 & 28 & 32 \end{bmatrix}.$$

2. **vSIS assumption.**

The (vanishing-)vSIS assumption states that, for a random row-tensor $\mathbf{F} \leftarrow_\$ \mathcal{R}_q^{n\times d^{\otimes\mu}}$, it is hard to find a short vector $\mathbf{x}$ satisfying $\mathbf{F}\mathbf{x} = \mathbf{0} \bmod q$.
Based on this assumption, for a short vector $\mathbf{w}$, the vector $\mathbf{y} = \mathbf{F}\mathbf{w} \bmod q$ is a computationally binding commitment (similar to Ajtai commitment).

Depending on the tensor structure of $\mathbf{F}$, the product $\mathbf{F}\mathbf{w}$ can be seen as the evaluation of a polynomial with coefficients given by $\mathbf{w}$.

- For example, if $m = 2^\mu$ and $\mathbf{F} = (1, f, \ldots, f^{m-1}) = (1, f^{m/2}) \otimes (1, f^{m/4}) \otimes \ldots \otimes (1, f)$, then $\mathbf{F}\mathbf{w} = \sum_{i=0}^{m-1} w_i f^i$.

3. **Principal linear relation.**

$$\Xi^{\mathsf{lin}}_{\hat{n},n,m,r,\beta} := \begin{cases} ((\mathbf{H},\mathbf{F},\mathbf{Y}),\mathbf{W}) \\ \mathbf{H} \in \mathcal{R}_q^{\hat{n}\times n} \quad \mathbf{F} \in \mathcal{R}_q^{n\times m} \\ \mathbf{Y} \in \mathcal{R}_q^{\hat{n}\times r} \quad \mathbf{W} \in \mathcal{R}^{m\times r} \\ \|\mathbf{W}\|_{\sigma,2} \le \beta \quad \text{and} \quad \mathbf{HFW} = \mathbf{Y} \bmod q \end{cases}$$

where $\mathbf{H}$ is restricted to take the form $\mathbf{H} = [\,\mathbf{I}_{\hat{n}} \quad \underline{\mathbf{H}}\,]$ for some $\hat{n} \le n$. Throughout, we think of $n = \bar{n} + \underline{n}$ and $\mathbf{F} = \begin{bmatrix} \bar{\mathbf{F}} \\ \underline{\mathbf{F}} \end{bmatrix} \in \mathcal{R}_q^{(\bar{n}+\underline{n})\times m}$ as consisting of a top part $\bar{\mathbf{F}} \in \mathcal{R}_q^{\bar{n}\times m}$ and a bottom part $\underline{\mathbf{F}} \in \mathcal{R}_q^{\underline{n}\times m}$.

- the top rows $\bar{\mathbf{F}}$ for the vSIS **commitment**;
- the bottom rows $\underline{\mathbf{F}}$ for **polynomial evaluations**.

Compare with the main relation in Latticefold+ [2] (Commit + Linear relation)

$$\mathcal{R}_{lin,B} = \left\{ (\mathbb{i},\mathbb{x},\mathbb{w}) \middle| \begin{array}{l} \mathbb{x} = (\mathsf{cm}_{\mathbf{f}}, \mathbf{r}, \mathbf{v}_1, \ldots, \mathbf{v}_{n_{lin}}), \\ \mathbb{w} = \mathbf{f}, \\ \mathsf{cm}_{\mathbf{f}} = \mathsf{Com}(\mathbf{f}), \\ \|\mathbf{f}\|_\infty < B, \\ \forall i \in [n_{lin}],\ \langle \mathbf{M}^{(i)}\mathbf{f}, \mathsf{tensor}(\mathbf{r})\rangle = \mathbf{v}_i \end{array} \right\}$$

2. Folding summary

A brief summary of the functionalities of these RoKs:

1. Π^{join} joins two claims together.
2. Π^{norm} implements an exact norm-check.
3. Π^{RP} and $\Pi^{\otimes\mathsf{RP}}$ are approximate norm checks via (un)structured random projections.
4. Π^{fold} folds $\mathbf{W} \in \mathcal{R}^{m\times r}$ and reduces r to 1, i.e. $\mathbf{W} \to \mathbf{w} \in \mathcal{R}^m$
5. $\Pi^{b\text{-decomp}}$ b-ary decomposes $\mathbf{W}$, reducing the norm bound β but increasing the witness dimension r.

The composition of the RoKs for the folding scheme consists of the following six individual RoKs:

$$\Pi^{\mathsf{join}} \to \Pi^{\mathsf{norm}} \to \Pi^{\otimes\mathsf{RP}} \to \underset{\to}{\Pi^{\mathsf{fold}}} \to \Pi^{\mathsf{join}} \to \Pi^{\mathsf{batch}} \to \Pi^{b\text{-decomp}}.$$

The chain of the reduction is prefixed with an additional Π^{join} RoK, which merges the multiple instances into a single relation.

- Start with the relation $\Xi^{\mathsf{lin}}_{\hat{n},n,\mu,r_{\mathsf{acc}}+L,\beta}$ where L is the number of instances to be folded.
- Then norm check
  - Precise norm check: The first RoK Π^{norm} takes as input the relation $\Xi^{\mathsf{lin}}_{\hat{n},n,\mu,r_{\mathsf{acc}}+L,\beta}$ and outputs the relation $\Xi^{\mathsf{lin}}_{\hat{n}+2,n+2,\mu,r_{\mathsf{acc}}+L,\beta}$.

- Approximate norm check: Next, the protocol $\Pi^{\otimes\mathsf{RP}}$ takes as input the relation $\Xi^{\mathsf{lin}}_{\hat{n}+2,n+2,\mu,r_{\mathsf{acc}}+L,\beta}$ and outputs two relations: $\Xi^{\mathsf{lin}}_{\hat{n}+3,n+3,m,r_{\mathsf{acc}}+L,\beta}$ and $\Xi^{\mathsf{lin}}_{\bar{n}+1,\bar{n}+1,m,1,\hat{\beta}}$ where $\hat{\beta} = m_{\mathsf{rp}} \cdot \beta$ and $m_{\mathsf{rp}} = \mathcal{O}(\lambda)$.

- Next fold
  - The next RoK Π^{fold} takes the first relation $\Xi^{\mathsf{lin}}_{\hat{n}+3,n+3,m,r_{\mathsf{acc}}+L,\beta}$ and outputs the relation $\Xi^{\mathsf{lin}}_{\hat{n}+3,n+3,m,1,(r_{\mathsf{acc}}+L)\gamma\beta}$ where γ is an expansion factor of the challenge set.
  - Two branches are then merged using the RoK Π^{join}, resulting in the relation $\Xi^{\mathsf{lin}}_{\hat{n}+4,n+4,m,2,\max(\hat{\beta},(r_{\mathsf{acc}}+L)\gamma\beta)}$.
- Last decompose
  - Eventually, the last two RoKs Π^{batch} and $\Pi^{b\text{-decomp}}$ take as input the relation $\Xi^{\mathsf{lin}}_{\hat{n}+4,n+4,m,2,\max(\hat{\beta},(r_{\mathsf{acc}}+L)\gamma\beta)}$ and output the relation $\Xi^{\mathsf{lin}}_{\hat{n},n+4,m,2\ell,\beta}$ as ℓ is set so that decomposition of the witness of norm bounded by $\max(\hat{\beta},(r_{\mathsf{acc}}+L)\gamma\beta)$ results in a witness of norm bounded by β.

3. The protocols

3.1 Π^{join} : merge many instances into one [4]

- input: L instances of relation $\Xi^{\mathsf{lin}}_{\hat{n},n,m,1,\beta}$ and one relation $\Xi^{\mathsf{lin}}_{\hat{n},n,m,r_{acc},\beta}$ (Old folded relations)
- output: relation $\Xi^{\mathsf{lin}}_{\hat{n},n,m,r_{\mathsf{acc}}+L,\beta}$
  This requires sending cross-terms of the bottom rows of $\mathbf{F}$ corresponding to different instances.
  As a result, the number of rows of $\mathbf{F}$ increases by $(n-\bar{n})\cdot(L-1)$ where L is the number of instances to be folded.

$$\Pi^{\mathsf{join}} : \Xi^{\mathsf{lin}}_{m,\hat{n}_0,r_0,\beta} \times \Xi^{\mathsf{lin}}_{m,\hat{n}_1,r_1,\beta} \rightarrow \Xi^{\mathsf{lin}}_{m,\hat{n}_0+\hat{n}_1-\overline{n},r_0+r_1,\beta}$$

$$\mathcal{P}\begin{pmatrix}((\mathbf{H}_0,\mathbf{F}_0,\mathbf{Y}_0),\mathbf{W}_0)\\((\mathbf{H}_1,\mathbf{F}_1,\mathbf{Y}_1),\mathbf{W}_1)\\ \text{s.t. } \overline{\mathbf{F}}_0=\overline{\mathbf{F}}_1=\overline{\mathbf{F}}\end{pmatrix} \qquad \mathcal{V}\begin{pmatrix}(\mathbf{H}_0,\mathbf{F}_0,\mathbf{Y}_0)\\(\mathbf{H}_1,\mathbf{F}_1,\mathbf{Y}_1)\\ \text{s.t. } \overline{\mathbf{F}}_0=\overline{\mathbf{F}}_1=\overline{\mathbf{F}}\end{pmatrix}$$

$$1: \quad \underline{\mathbf{Y}}_{01} := \underline{\mathbf{H}}_0\underline{\mathbf{F}}_0\mathbf{W}_1 \bmod q$$

$$2: \quad \underline{\mathbf{Y}}_{10} := \underline{\mathbf{H}}_1\underline{\mathbf{F}}_1\mathbf{W}_0 \bmod q \quad \xrightarrow{\underline{\mathbf{Y}}_{01},\,\underline{\mathbf{Y}}_{10}}$$

$$3: \quad \mathbf{H} := \begin{pmatrix}\mathbf{I}_{\overline{n}} & & \\ & \underline{\mathbf{H}}_0 & \\ & & \underline{\mathbf{H}}_1\end{pmatrix} \quad \mathbf{F} := \begin{pmatrix}\overline{\mathbf{F}}\\ \underline{\mathbf{F}}_0\\ \underline{\mathbf{F}}_1\end{pmatrix} \quad \mathbf{Y} := \begin{pmatrix}\overline{\mathbf{Y}}_0 & \overline{\mathbf{Y}}_1\\ \underline{\mathbf{Y}}_0 & \underline{\mathbf{Y}}_{01}\\ \underline{\mathbf{Y}}_{10} & \underline{\mathbf{Y}}_1\end{pmatrix}$$

$$4: \quad ((\mathbf{H},\mathbf{F},\mathbf{Y}),(\mathbf{W}_0\ \mathbf{W}_1)) \stackrel{?}{\in} \Xi^{\mathsf{lin}}_{m,\hat{n}_0+\hat{n}_1-\overline{n},r_0+r_1,\beta}$$

Fig. 4: RoK Π^{join}.

3.2 Norm Check [3]

- input: relation $\Xi^{\mathsf{lin}}_{\hat{n},n,m,r_{\mathsf{acc}}+L,\beta}$

Π^{norm}: $\Xi^{\mathsf{norm}}_{\hat{n},n,\mu,r,\beta} \to \Xi^{\overline{\mathsf{sum}}}_{\hat{n},n,\mu,r,\beta}$

$\mathcal{P}((\mathbf{H},\mathbf{F},\mathbf{Y},\nu),\mathbf{W})$		$\mathcal{V}(\mathbf{H},\mathbf{F},\mathbf{Y},\nu)$
1 : **parse** $\mathbf{W} = (\mathbf{w}_i)_{i\in[r]}$		
2 : $\mathbf{t}^{\mathsf{T}} := (\langle \mathbf{w}_i, \overline{\mathbf{w}}_i\rangle)_{i\in[r]}$	$\xrightarrow{\mathbf{t}}$	$\mathbf{t}^{\mathsf{T}} := (t_0,\ldots,t_{r-1})$
3 :		**for** $i \in [r]$: $\mathsf{Trace}(t_i) \overset{?}{\leq} \nu^2$

4 : $((\mathbf{H},\mathbf{F},\mathbf{Y},\mathbf{t}),\mathbf{W}) \in \Xi^{\overline{\mathsf{sum}}}_{\hat{n},n,\mu,r,\beta}$

Π^{sum}: $\Xi^{\overline{\mathsf{sum}}}_{\hat{n},n,\mu,r,\beta} \to \Xi^{\mathsf{lde}\text{-}\otimes}_{\hat{n},n,\mu,r,\beta,2}$

$\mathcal{P}((\mathbf{H},\mathbf{F},\mathbf{Y},\mathbf{t}),\mathbf{W})$		$\mathcal{V}((\mathbf{H},\mathbf{F},\mathbf{Y},\mathbf{t}))$
1 :		$u \leftarrow\!\$\ \mathbb{F}_{q^e}^{\times}$
2 : $\tilde{f} := \mathbf{u}^{\mathsf{T}} \cdot \mathsf{CRT}(\mathsf{LDE}[\mathbf{W}] \odot \mathsf{LDE}[\overline{\mathbf{W}}]) \bmod q \in \mathbb{F}_{q^e}[\mathsf{x}^{\mu}]$	$\xleftarrow{\mathbf{u}^{\mathsf{T}} := (u^i)_{i\in[r\varphi/e]}}$	$a_0 := \mathbf{u}^{\mathsf{T}} \cdot \mathsf{CRT}(\mathbf{t}) \bmod q$
3 : **for** $j \in [\mu]$:		
4 : $g_j(\mathsf{x}) := \sum_{\mathbf{z}_j \in [d]^{\mu-j-1}} \tilde{f}(r_0,\ldots,r_{j-1},\mathsf{x},\mathbf{z}_j) \bmod q$	$\xrightarrow{g_j(\mathsf{x}) \in \mathbb{F}_{q^e}[\mathsf{x}]}$	$a_j \overset{?}{=} \sum_{z\in[d]} g_j(z) \bmod q$
5 : sumcheck for LDE	$\xleftarrow{r_j}$	$r_j \leftarrow \mathbb{F}_{q^e}^{\times}$
		$a_{j+1} := g_j(r_j) \bmod q$
6 : $\mathbf{r}^{\mathsf{T}} := (\mathsf{CRT}^{-1}(\mathbf{1}_{\varphi/e} \cdot r_j)^{\mathsf{T}})_{j\in[\tilde{\mu}]}$		
7 : $\mathbf{s}_0 := \mathsf{LDE}[\mathbf{W}](\mathbf{r}) \bmod q$		
8 : $\mathbf{s}_1 := \mathsf{LDE}[\mathbf{W}](\overline{\mathbf{r}}) \bmod q$	$\xrightarrow{(\mathbf{s}_0,\mathbf{s}_1) \in \mathcal{R}_q^{r\times 2}}$	$a_{\mu} \overset{?}{=} \mathbf{u}^{\mathsf{T}} \cdot \mathsf{CRT}(\mathbf{s}_0 \odot \overline{\mathbf{s}}_1) \bmod q$

9 : $((\mathbf{H},\mathbf{F},\mathbf{Y},(\mathbf{r}_i,\mathbf{s}_i)_{i\in[2]}),\mathbf{W}) \in \Xi^{\mathsf{lde}\text{-}\otimes}_{\hat{n},n,\mu,r,\beta,2}$ with $\mathbf{r}_0 := \mathbf{r}, \mathbf{r}_1 := \overline{\mathbf{r}}$

μ times

- output:

$$\Xi^{\mathrm{lde}(-\otimes)}_{\hat{n},n,\mu,\tilde{\mu},r,\beta,2} := \left\{ \begin{array}{l} ((\mathbf{H},\mathbf{F},\mathbf{Y},(\mathbf{r}_i,\mathbf{s}_i,\mathbf{M}_i)_{i\in[2]}),\mathbf{W}) : \\ \quad ((\mathbf{H},\mathbf{F},\mathbf{Y}),\mathbf{W}) \in \Xi^{\mathrm{lin}}_{\hat{n},n,\mu,r,\beta} \\ \quad (\mathbf{r}_i,\mathbf{s}_i,\mathbf{M}_i)_{i\in[2]} \in (\mathcal{R}_q^{\tilde{\mu}} \times \mathcal{R}_q^{r} \times \mathcal{R}_q^{\tilde{m}\times m})^2 \\ \quad \forall_{i\in[2]} \quad \mathbf{s}_i = \mathrm{LDE}[\mathbf{M}_i\mathbf{W}](\mathbf{r}_i) \bmod q \end{array} \right\}$$

By embedding $\mathbf{s_1},\mathbf{s_2}$ into $\mathbf{HFW}=\mathbf{Y}$, we get the relation:

$$\Xi^{\mathsf{lin}}_{\hat{n}+2,n+2,m,r_{\mathsf{acc}}+L,\beta}.$$

3.3 $\Pi^{\otimes\mathsf{RP}}$: Approximate range check [5]

3.3.1 Previous results

1. **subtractive sets**. A set $C \subset \mathcal{R}$ is called subtractive if for any two distinct elements $c, c' \in C$, the inverse of $c - c'$ exists in the ring and has low norm.

- subtractive sets in knowledge soundness
  after extracting two accepting transcripts with challenges $c \neq c'$, one often recovers the witness in the form

$$\mathbf{w} := \frac{\mathbf{z} - \mathbf{z}'}{c - c'},$$

  where $\mathbf{z}, \mathbf{z}'$ are short vectors from the transcripts.
- The soundness error of this methodology is essentially $1/|C|$, so ideally one would like C to be very large.
- However, known subtractive sets in the relevant rings have only polynomial size in the security parameter. As a result, one needs many repetitions to drive the soundness error down to negligible, which increases the proof size.

2. **Johnson--Lindenstrauss Random Projection**
   - Motivation: To use an exponential-size unstructured challenge set and get one-shot soundness, reducing r to $O(1)$ and dropping communication to $O(1)$ ring elements per round.
   - Johnson--Lindenstrauss Random Projection
     the verifier samples a random small integer matrix $\mathbf{J}$, say $\{-1, 0, 1\}$, and the prover returns the projected image

$$\mathbf{v} := \mathbf{J}\,\mathsf{cf}(\mathbf{w}) \bmod q,$$

     where $\mathrm{cf}(\mathbf{w})$ is the coefficient vector of the ring elements.
     The verifier then checks that $\mathbf{v}$ has low norm, and the prover proves the well-formedness of the above equation.
     - The correctness: with high probability over the choice of $\mathbf{J}$, the projection approximately preserves norm, so $\|\mathbf{v}\| \approx \|\mathbf{w}\|$.
     - Knowledge soundness, if an extracted candidate witness actually had large norm, then except with exponentially small probability in the projection dimension, its projection would also have large norm.

3.3.2 The protocol $\otimes$RP

$\otimes$RP $\rightarrow$ apply tensor production on the random project.

Setup: The prover holds a witness $\mathbf{w}$ satisfying $\mathbf{F}\mathbf{w} = \mathbf{y}$ with $\|\mathbf{w}\| \leq \beta$.

Step 1 — Sample the projection matrix.
The verifier samples a random matrix $\mathbf{J}' \in \mathcal{R}^{n_{\mathsf{rp}} \times m_{\mathsf{rp}}}$ with entries from some small integer distribution χ . The full projection is block-diagonal: $\mathbf{J} = \mathbf{I} \otimes \mathbf{J}'$.

Step 2 — Compute the projected witness.
The prover computes $\hat{\mathbf{w}} := \mathbf{J}\mathbf{w} \bmod q$, st. $||\mathbf{w}|| \approx ||\mathbf{J}\mathbf{w}||$,then the prover

- sends out $\hat{\mathbf{w}}$ directly, the verifier checks its norm.
- Alternative: commits to $\hat{\mathbf{w}}$ to reduce its size: $\hat{\mathbf{F}}\hat{\mathbf{w}} = \hat{\mathbf{z}}$.

Step 3 — Produce two linked claims.

$$\underbrace{\begin{bmatrix}\tilde{\mathbf{F}}\\ \mathbf{I}\otimes\mathbf{J}'\end{bmatrix}\mathbf{w}=\begin{bmatrix}\mathbf{y}\\ \hat{\mathbf{w}}\end{bmatrix}}_{\text{claim on original }\mathbf{w}}\qquad \underbrace{\begin{bmatrix}\hat{\mathbf{F}}\\ \mathbf{I}\end{bmatrix}\hat{\mathbf{w}}=\begin{bmatrix}\hat{\mathbf{z}}\\ \mathbf{w}\end{bmatrix}}_{\text{claim on projected }\tilde{\mathbf{w}}}$$

The shared $\mathbf{w}$ cryptographically ties the two together — a cheating prover cannot swap in a different $\tilde{\mathbf{w}}$ without breaking both claims simultaneously.

$\Pi^{\otimes\mathsf{RP}}: \Xi^{\mathsf{lin}\text{-}\otimes(r,m')}_{m,\hat{n},r,\beta} \to \Xi^{\mathsf{lin}\text{-}\otimes(r,m')}_{m,\hat{n}+1,r,\beta} \times \Xi^{\mathsf{lin}\text{-}\otimes(r,m')}_{m,\overline{n}+1,1,\hat{\beta}}$ where $m_{\mathsf{rp}}/n_{\mathsf{rp}} = r, m' := m/r$

$\mathcal{P}((\mathbf{H},\mathbf{F},\mathbf{Y}),\mathbf{W})$		$\mathcal{V}(\mathbf{H},\mathbf{F},\mathbf{Y})$
1 : $\widehat{\mathbf{J}} := \mathbf{I}_{m/m_{\mathsf{rp}}} \otimes \mathbf{J} \in \mathcal{R}_q^{m'\times m}$	$\xleftarrow{\mathbf{J}}$	$\mathbf{J} \leftarrow \chi_{\mathsf{ghl}} \sim \{-1,0,1\}^{n_{\mathsf{rp}}\times m_{\mathsf{rp}}}$
2 : $\widehat{\mathbf{W}} := \widehat{\mathbf{J}}\mathbf{W} \in \mathcal{R}_q^{m'\times r}$		
3 : $\widehat{\mathbf{w}} := \mathsf{vec}(\widehat{\mathbf{W}}) \in \mathcal{R}_q^{m}$		
4 : $\overline{\mathbf{z}} := \overline{\mathbf{F}}\widehat{\mathbf{w}} \bmod q \in \mathcal{R}_q^{\overline{n}}$	$\xrightarrow{\overline{\mathbf{z}}}$	$c \leftarrow \mathcal{R}_q^{\times}$
5 : $c_0 := c^{m'}, \quad c_1 := c$	$\xleftarrow{c}$	
6 : $\mathbf{c}_0 := (1, c_0, \ldots, c_0^{r-1})^{\mathsf{T}}$		
7 : $\mathbf{c}_1 := (1, c_1, \ldots, c_1^{m'-1})^{\mathsf{T}}$		
8 : $\mathbf{c} := (1, c, \ldots, c^{m-1})^{\mathsf{T}}$ // $\mathbf{c} = \mathbf{c}_0 \otimes \mathbf{c}_1$		
9 : $\mathbf{r}^{\mathsf{T}} := \mathbf{c}_1^{\mathsf{T}}\widehat{\mathbf{W}} \bmod q \in \mathcal{R}_q^{r}$	$\xrightarrow{\mathbf{r}}$	

10 : $\widetilde{\mathbf{H}} := \begin{pmatrix}\mathbf{H} & \mathbf{0}\\ \mathbf{0} & 1\end{pmatrix} \in \mathcal{R}_q^{(\hat{n}+1)\times(n+1)}$ $\quad \widetilde{\mathbf{F}} := \begin{pmatrix}\mathbf{F}\\ \mathbf{c}_1^{\mathsf{T}}\widehat{\mathbf{J}}\end{pmatrix} \in \mathcal{R}_q^{(n+1)\times m}$ $\quad \widetilde{\mathbf{Y}} := \begin{pmatrix}\mathbf{Y}\\ \mathbf{r}^{\mathsf{T}}\end{pmatrix} \in \mathcal{R}_q^{(\hat{n}+1)\times r}$

$\widehat{\mathbf{F}} := \begin{pmatrix}\overline{\mathbf{F}}\\ \mathbf{c}^{\mathsf{T}}\end{pmatrix} \in \mathcal{R}_q^{(\overline{n}+1)\times m'}$ $\quad \widehat{\mathbf{y}} := \begin{pmatrix}\overline{\mathbf{z}}\\ \mathbf{c}_0^{\mathsf{T}}\cdot\mathbf{r}\end{pmatrix} \in \mathcal{R}_q^{\overline{n}+1}$

11 : $((\widetilde{\mathbf{H}},\widetilde{\mathbf{F}},\widetilde{\mathbf{Y}}),\mathbf{W}) \in \Xi^{\mathsf{lin}\text{-}\otimes(r,m')}_{m,\hat{n}+1,r,\beta}$ and $((\mathbf{I},\widehat{\mathbf{F}},\widehat{\mathbf{y}}),\widehat{\mathbf{w}}) \in \Xi^{\mathsf{lin}\text{-}\otimes(r,m')}_{m,\overline{n}+1,1,\hat{\beta}}$

Fig. 3: RoK $\Pi^{\otimes\mathsf{RP}}$.

- input: relation $\Xi^{\mathsf{lin}}_{\hat{n}+2,n+2,m,r_{\mathsf{acc}}+L,\beta}$.
- output: $\Xi^{\mathsf{lin}}_{\hat{n}+3,n+3,m,r_{\mathsf{acc}}+L,\beta}$ and $\Xi^{\mathsf{lin}}_{\bar{n}+1,\bar{n}+1,m,1,\hat{\beta}}$ where $\hat{\beta} = m_{\mathsf{rp}}\cdot\beta$ and $m_{\mathsf{rp}} = \mathcal{O}(\lambda)$.

3.4 Fold

3.4.1 Fold the witness [4]

- input: the first relation $\Xi^{\mathsf{lin}}_{\hat{n}+3,n+3,m,r_{\mathsf{acc}}+L,\beta}$
- outputs the relation $\Xi^{\mathsf{lin}}_{\hat{n}+3,n+3,m,1,(r_{\mathsf{acc}}+L)\gamma\beta}$

The fold protocol Π^{fold} reduces the witness width r to 1 (by random linear combining the columns).

$\mathcal{P}((\mathbf{H},\mathbf{F},\mathbf{Y}),\mathbf{W})$		$\mathcal{V}((\mathbf{H},\mathbf{F},\mathbf{Y}))$
	$\xleftarrow{\quad\mathbf{C}\quad}$	$\mathbf{C} \leftarrow\$\ \mathcal{C}_{\mathcal{R}} \in \mathcal{R}^{r_{\mathsf{in}} \times r_{\mathsf{out}}}$
$\widetilde{\mathbf{W}} := \mathbf{W} \cdot \mathbf{C}$		$\widetilde{\mathbf{Y}} := \mathbf{Y} \cdot \mathbf{C}$
	$\xrightarrow{\quad\boxed{\widetilde{\mathbf{W}}}\quad}$	$\boxed{((\mathbf{H},\mathbf{F},\widetilde{\mathbf{Y}}),\widetilde{\mathbf{W}}) \overset{?}{\in} \Xi^{\mathsf{lin}}_{m,r_{\mathsf{out}},\beta=\sqrt{r_{\mathsf{out}}}r_{\mathsf{in}}\gamma r_{\mathsf{in}}\beta}}$

Fig. 5. Protocol Π^{fold} folds instance of $\Xi^{\mathsf{lin}}_{\mathsf{par\text{-}in}}$ into $\Xi^{\mathsf{lin}}_{\mathsf{par\text{-}out}}$ with $\mathsf{par\text{-}in}$, $\mathsf{par\text{-}out}$ specified in Lemma 6. Π^{fold} sends the marked parts only as a *proof* (but not *reduction*) of knowledge.

3.4.2 Merge the relations from Π^{RP}, Π^{Fold}

- input: the relation $\Xi^{\mathsf{lin}}_{\bar{n}+1,\bar{n}+1,m,1,\hat{\beta}}$ from $\otimes$RP, and the relation $\Xi^{\mathsf{lin}}_{\hat{n}+3,n+3,m,1,(r_{\mathsf{acc}}+L)\gamma\beta}$ from Fold.
- output: $\Xi^{\mathsf{lin}}_{\hat{n}+4,n+4,m,2,\max(\hat{\beta},(r_{\mathsf{acc}}+L)\gamma\beta)}$
- $\bar{\mathbf{F}}$ (dimension $\bar{n}$) is shared, thus $\mathbf{H} \in \mathcal{R}^{(\hat{n}+4)\times(n+4)}$ and $\mathbf{F} \in \mathcal{R}^{(n+4)\times m}$.

$$\Pi^{\mathsf{join}} : \Xi^{\mathsf{lin}}_{m,\hat{n}_0,r_0,\beta} \times \Xi^{\mathsf{lin}}_{m,\hat{n}_1,r_1,\beta} \to \Xi^{\mathsf{lin}}_{m,\hat{n}_0+\hat{n}_1-\bar{n},r_0+r_1,\beta}$$

$$\mathcal{P}\begin{pmatrix}((\mathbf{H}_0,\mathbf{F}_0,\mathbf{Y}_0),\mathbf{W}_0)\\ ((\mathbf{H}_1,\mathbf{F}_1,\mathbf{Y}_1),\mathbf{W}_1)\\ \text{s.t.}\ \ \overline{\mathbf{F}}_0=\overline{\mathbf{F}}_1=\overline{\mathbf{F}}\end{pmatrix} \qquad \mathcal{V}\begin{pmatrix}(\mathbf{H}_0,\mathbf{F}_0,\mathbf{Y}_0)\\ (\mathbf{H}_1,\mathbf{F}_1,\mathbf{Y}_1)\\ \text{s.t.}\ \ \overline{\mathbf{F}}_0=\overline{\mathbf{F}}_1=\overline{\mathbf{F}}\end{pmatrix}$$

$1:\quad \underline{\mathbf{Y}}_{01} := \underline{\mathbf{H}}_0\underline{\mathbf{F}}_0\mathbf{W}_1 \bmod q$

$2:\quad \underline{\mathbf{Y}}_{10} := \underline{\mathbf{H}}_1\underline{\mathbf{F}}_1\mathbf{W}_0 \bmod q \qquad \xrightarrow{\quad\underline{\mathbf{Y}}_{01},\,\underline{\mathbf{Y}}_{10}\quad}$

$$3:\quad \mathbf{H} := \begin{pmatrix}\mathbf{I}_{\overline{n}} & & \\ & \underline{\mathbf{H}}_0 & \\ & & \underline{\mathbf{H}}_1\end{pmatrix} \quad \mathbf{F} := \begin{pmatrix}\overline{\mathbf{F}}\\ \underline{\mathbf{F}}_0\\ \underline{\mathbf{F}}_1\end{pmatrix} \quad \mathbf{Y} := \begin{pmatrix}\overline{\mathbf{Y}}_0 & \overline{\mathbf{Y}}_1\\ \underline{\mathbf{Y}}_0 & \underline{\mathbf{Y}}_{01}\\ \underline{\mathbf{Y}}_{10} & \underline{\mathbf{Y}}_1\end{pmatrix}$$

$$4:\quad ((\mathbf{H},\mathbf{F},\mathbf{Y}),(\mathbf{W}_0\ \mathbf{W}_1)) \overset{?}{\in} \Xi^{\mathsf{lin}}_{m,\hat{n}_0+\hat{n}_1-\overline{n},r_0+r_1,\beta}$$

Fig. 4: RoK Π^{join}.

3.5.1 Batch the statement [4]

- Input: $\Xi^{\mathsf{lin}}_{\hat{n}+4,n+4,m,2,\max(\hat{\beta},(r_{\mathsf{acc}}+L)\gamma\beta)}$
- Output: $\Xi^{\mathsf{lin}}_{\hat{n},n+4,m,2,\max(\hat{\beta},(r_{\mathsf{acc}}+L)\gamma\beta)}$
  - in the following figure, the challenge $\mathbf{c}^\top$ is a row, thus $\mathbf{H} \in \mathcal{R}^{(\hat{n}+4)\times(n+4)}$ is reduced to $\tilde{H} \in \mathcal{R}^{(\bar{n}+1)\times(n+4)}$ (top-part + 1-row)
  - pick $\mathbf{c}^\top$ to be $(\hat{n}-\bar{n})$-row to get $\mathbf{H} \in \mathcal{R}^{\hat{n}\times(n+4)}$, i.e., our output relation.

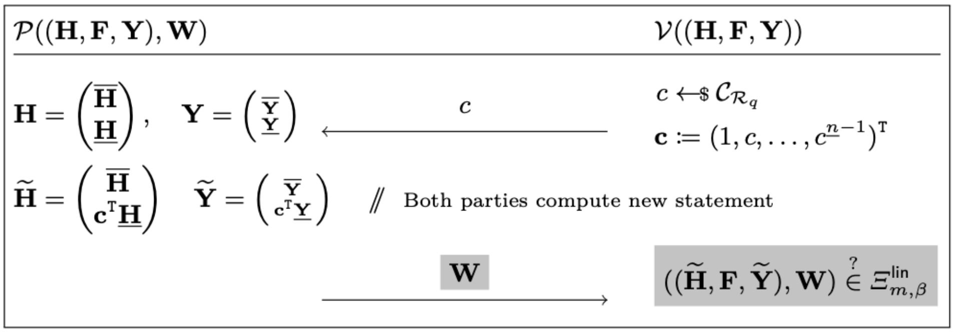

Fig. 6. Protocol Π^{batch} reduces an instance of $\Xi^{\mathsf{lin}}_{\mathsf{par\text{-}in}}$ to $\Xi^{\mathsf{lin}}_{\mathsf{par\text{-}out}}$ with $\mathsf{par\text{-}in}$, $\mathsf{par\text{-}out}$ specified in Lemma 4, with fewer rows by batching to the last $\underline{n}$ of $\mathbf{H}$. Π^{batch} sends the marked parts only as a *proof* (but not *reduction*) of knowledge.

3.5.2 b-ary Decomposition [4]

The b-ary decomposition protocol $\Pi^{b\text{-decomp}}$ reduces the witness norm β by b-ary decomposing the matrix $\mathbf{W}$ as $\sum_{i=0}^{\ell-1} b^i \mathbf{V}_i$, at the expense of increased width $\tilde{r}$ in the resulting $\widetilde{\mathbf{W}}$.

The prover needs to communicate $\mathbf{Y}_i = \mathbf{HFV}_i$.

The new witness is $\widetilde{\mathbf{W}} = (\mathbf{V}_0, \ldots, \mathbf{V}_{\ell-1})$ and the resulting $\widetilde{\mathbf{Y}}$ is $(\mathbf{Y}_0, \ldots, \mathbf{Y}_{\ell-1})$.

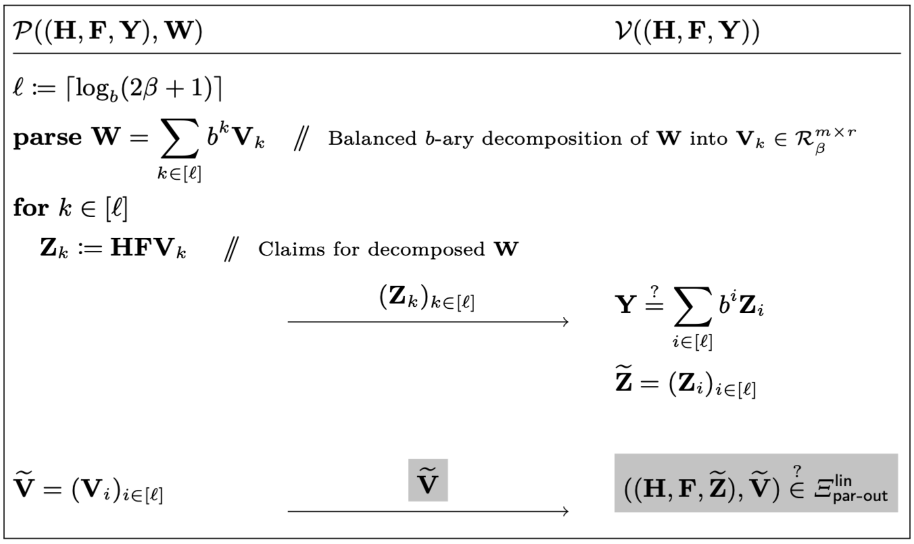

Fig. 3. Protocol $\Pi^{b\text{-decomp}}$, a reduction of $\Xi^{\mathsf{lin}}_{\mathsf{par\text{-}in}}$ to $\Xi^{\mathsf{lin}}_{\mathsf{par\text{-}out}}$ with $\mathsf{par\text{-}in}$, $\mathsf{par\text{-}out}$ specified in Lemma 4. As a *proof* (but not *reduction*) of knowledge, $\Pi^{b\text{-decomp}}$ sends the marked parts.

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